AI/ML Research Proposals

Emerging, Research-Driven Topics with Publication Potential

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How novelty is framed

The proposals are deliberately not presented as guaranteed "first-ever" ideas. Instead, novelty is framed as a specific research gap that is distinguishable from the closest identified prior work. This is important because several of these fields are moving rapidly in 2025–2026. Before journal submission, a final systematic search across Scopus, Web of Science, IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, ACL Anthology, and arXiv should be performed to update the related-work section and verify the exact novelty claim.

Proposal 1 Hierarchical Context Compression for Relational Foundation Models

1.1 Proposed Title

Hierarchical Context Compression for Efficient and Information-Preserving Relational Foundation Models

1.2 One-Line Summary

A learning-based framework that compresses extremely large relational neighborhoods while preserving the information that matters most for downstream prediction, reducing context size, latency, and compute without sacrificing accuracy.

1.3 Keywords

Relational foundation models; relational deep learning; context compression; tabular data; relational databases; representation learning; efficient AI; zero-shot learning; temporal aggregation; high-cardinality data.

1.4 Abstract

Relational data is fundamental to domains such as finance, healthcare, e-commerce, cybersecurity, and enterprise analytics, where useful information is distributed across interconnected tables rather than isolated records. Recent relational foundation models have demonstrated that a single pretrained model can transfer across heterogeneous relational datasets and downstream tasks. However, real-world entities may be associated with thousands or millions of historical records, creating a severe context-scaling problem. Existing strategies based on fixed neighborhood budgets, truncation, sampling, or manually designed aggregation can discard task-relevant information while increasing inference cost. This research proposes a hierarchical context-compression framework that dynamically preserves task-relevant information from large relational neighborhoods while substantially reducing the context presented to a relational foundation model.

The proposed method will combine relational structure, temporal information, semantic attributes, and learned importance scores to construct multi-level representations of relational neighborhoods. Instead of treating all records equally, the framework will selectively retain informative observations while summarizing less informative records into compact representations. The study will investigate the trade-off between context size, predictive accuracy, inference

latency, and memory consumption across multiple relational benchmark datasets and downstream tasks. Comparisons will be made against full-context inference, truncation, random sampling, fixed aggregation, and existing relational-model baselines.

The expected contribution is a principled method for scalable relational inference together with an empirical characterization of how much relational context is actually necessary for reliable prediction. The work aims to make relational foundation models more practical for high-cardinality real-world databases and to provide a reproducible benchmark for measuring the accuracy–efficiency trade-off created by context compression.

1.5 Prior Work

The Relational Transformer introduced a foundation-model architecture for relational data, using schema-aware tokenization and relational attention and demonstrating zero-shot transfer across unseen relational tasks [1]. RDB-PFN subsequently showed that relational foundation models can be pretrained entirely from synthetic relational databases and tasks, addressing data scarcity for relational pretraining [2]. More recently, work on context-window failures reported severe degradation when high-cardinality entities force relational models to truncate large neighborhoods; on a synthetic financial benchmark, temporal pre-aggregation substantially improved performance [3]. These studies strongly establish the importance of the problem but do not fully solve the question of *learned, task-aware, hierarchical information preservation* across varying neighborhood cardinalities.

1.6 Research Gap and Proposed Novelty

The proposed research does not merely shorten the context. Its central hypothesis is that context reduction should be treated as a learned information-allocation problem. The system will learn which observations deserve exact preservation and which can be safely compressed, conditioned on the downstream task. The novelty therefore lies in the combination of: (i) hierarchical relational compression, (ii) task-aware information selection, and (iii) explicit accuracy–latency–memory analysis across multiple relational datasets. The contribution can be formalized as a Pareto trade-off between predictive utility and computational resource usage.

1.7 Objectives

1. Quantify context-window failures and computational bottlenecks in high-cardinality relational learning.
2. Develop a hierarchical, task-aware representation of large relational neighborhoods.
3. Compare learned compression with truncation, random sampling, and fixed temporal/statistical aggregation.
4. Measure predictive accuracy, memory use, inference latency, and context reduction jointly.
5. Release a reproducible implementation and benchmark protocol for relational context efficiency.

1.8 Research Timeline

Period	Planned Work
Month 1	Literature review; reproduce selected relational foundation-model baselines; finalize datasets, tasks, and evaluation metrics.
Month 2	Characterize context-window failures; implement truncation, sampling, and fixed aggregation baselines.
Months 3–4	Design and implement the hierarchical task-aware compression mechanism; conduct initial experiments and ablations.
Month 5	Large-scale cross-dataset experiments; analyze accuracy, context reduction, memory, and inference latency.
Month 6	Generalization and robustness analysis; finalize reproducibility package, report, and manuscript for journal submission.

Proposal 2 Risk-Aware Synthetic Tabular Data for Reliable ML under Distribution Shift

2.1 Proposed Title

Risk-Aware Synthetic Tabular Data Generation for Reliable Machine Learning under Distribution Shift

2.2 One-Line Summary

A synthetic-data framework that optimizes not only statistical fidelity and downstream accuracy, but also calibration, privacy, fairness, and reliability when the deployment distribution changes.

2.3 Keywords

Synthetic data; tabular data; diffusion models; generative models; distribution shift; out-of-distribution generalization; calibration; uncertainty; privacy; fairness; robust machine learning.

2.4 Abstract

Synthetic tabular data has become an important approach for addressing data scarcity, privacy restrictions, class imbalance, and limited access to sensitive datasets. Current evaluation practices, however, often emphasize statistical similarity to real data and downstream performance under conditions that resemble the training distribution. A synthetic dataset can match marginal statistics and still produce downstream models that are poorly calibrated or unreliable when deployed under temporal, demographic, feature, or missingness shifts. This research proposes a risk-aware framework in which the quality of synthetic data is assessed and optimized according to downstream reliability under distribution shift rather than fidelity alone.

The proposed framework will evaluate synthetic datasets using multiple dimensions including statistical fidelity, downstream utility, privacy leakage, fairness, uncertainty calibration, and out-of-distribution robustness. Controlled shifts will be introduced across temporal distributions, feature marginals, subgroup proportions, class prevalence, and missing-data patterns. Multiple synthetic-data paradigms, including GAN-based, variational, and diffusion-based generators,

will be evaluated under a common protocol. A risk-aware objective will then be investigated in which the generation process is encouraged to preserve the properties most relevant to reliable downstream decisions.

The expected contribution is a deployment-oriented evaluation and optimization framework that moves synthetic-data research beyond the question "Do the generated rows look real?" toward the more consequential question "Do models trained on these rows remain reliable when real-world conditions change?" The study will produce a reproducible benchmark and quantitative analysis of the trade-offs between fidelity, utility, privacy, fairness, and robustness, providing a practical basis for selecting synthetic data for safety-sensitive machine-learning applications.

2.5 Prior Work

Recent surveys show that tabular synthesis now spans GANs, VAEs, energy-based approaches, diffusion models, and LLM-based generators, with recurring challenges around fidelity, utility, privacy, diversity, and evaluation [4, 5]. A large benchmarking study in *Neurocomputing* evaluated five model families across 16 diverse datasets and found that careful dataset-specific tuning can substantially affect synthetic-data quality [6]. A 2025 systematic review further highlighted the lack of consensus in synthetic-health-data evaluation, inconsistent metrics, limited domain-expert involvement, and reproducibility weaknesses [7]. These studies motivate a stronger deployment-centered evaluation framework rather than another generator-only contribution.

2.6 Research Gap and Proposed Novelty

The proposed work focuses on the gap between synthetic-data fidelity and deployment reliability. Instead of treating out-of-distribution behavior as a separate downstream experiment, the study will make downstream robustness and calibration part of the evaluation and potentially part of the optimization objective. The key novelty is a unified risk-aware analysis linking synthetic-data properties to the reliability of decisions made by models trained on the synthetic data. This enables a direct study of which synthesis properties matter under realistic distribution shift.

2.7 Objectives

1. Build a unified evaluation framework for synthetic tabular data across fidelity, utility, privacy, fairness, and robustness.
2. Construct controlled but realistic distribution-shift scenarios for downstream evaluation.
3. Compare state-of-the-art synthetic-data families under identical experimental conditions.
4. Develop and test a risk-aware objective that rewards downstream reliability rather than fidelity alone.
5. Quantify the relationship between synthetic-data quality metrics and real deployment reliability.

2.8 Research Timeline

Period	Planned Work
Month 1	Literature review; select datasets and representative GAN, VAE, and diffusion baselines; define evaluation protocol.
Month 2	Implement fidelity, utility, privacy, fairness, calibration, and OOD metrics; construct controlled distribution-shift scenarios.
Months 3–4	Develop the risk-aware scoring/objective framework and integrate it with selected synthesis pipelines.
Month 5	Conduct cross-dataset experiments, ablations, and statistical significance analysis; identify failure cases.
Month 6	Consolidate results, prepare the open benchmark and manuscript, and formulate practical synthetic-data selection guidelines.

Proposal 3 Repair-Safe AI Code Generation

3.1 Proposed Title

Repair-Safe AI Code Generation: Preventing Security Regression in Automated Vulnerability Remediation

3.2 One-Line Summary

A verification-driven AI coding pipeline that accepts an automated security patch only when it improves security while preserving functionality and avoiding newly introduced vulnerabilities.

3.3 Keywords

AI-generated code; large language models; automated program repair; secure code generation; vulnerability detection; security regression; static analysis; program verification; software security.

3.4 Abstract

Large language models are increasingly used for software development tasks including code generation, debugging, vulnerability detection, and automated program repair. Although current systems can identify and fix many security weaknesses, an automatically generated fix cannot be assumed to be safe: it may break intended functionality, fail to remove the root cause, or introduce a different vulnerability. This creates a security and reliability barrier to the deployment of AI-assisted software engineering, especially in security-sensitive applications. This research proposes a repair-safe code-generation framework that verifies an automatically generated vulnerability fix before accepting the patch.

The proposed framework will implement a generate–verify–repair loop. A language model will produce multiple candidate patches, while independent verification components will evaluate security and functional correctness using static analysis, vulnerability detectors, unit or regression tests, semantic comparison, and security-regression analysis. Instead of optimizing only for vulnerability removal, the proposed system will consider four competing objectives: security improvement, functional preservation, regression avoidance, and minimal code modification. A

confidence-based acceptance policy will reject potentially unsafe patches and request additional candidate generation or verification when evidence is insufficient.

Evaluation will use public vulnerability datasets and real-world software repositories. Primary metrics will include vulnerability-removal rate, newly introduced vulnerabilities, functional test preservation, repair success rate, patch size, and verification cost. The expected contribution is a practical methodology for shifting AI-assisted program repair from "generate a patch" toward "generate, verify, and safely accept a patch." This has direct relevance to secure software engineering and could support safer adoption of generative AI in real development workflows.

3.5 Prior Work

Earlier work established that LLMs can detect, classify, and repair vulnerabilities in generated code, but with substantial limitations [8]. Recent empirical work using the real-world DevGPT dataset evaluated modern LLMs for detection and repair and reported substantial improvement over earlier generations, while also warning that LLM-generated code remains vulnerable and that confident model outputs can be incorrect [9]. A 2026 automated remediation pipeline combined CodeQL, Bandit, an independent validator, threat-context enrichment, and LLM-based fixing; importantly, it reported that remediation introduced new vulnerabilities in approximately 15–22% of tested cases [10]. These findings directly expose the limitation that motivates this proposal: successful repair of the original vulnerability is not equivalent to a security-safe patch.

3.6 Research Gap and Proposed Novelty

The central research gap is the lack of a principled, model-independent acceptance layer for AI-generated security repairs. Rather than proposing yet another vulnerability detector or another code-generation model, this work treats patch acceptance as a verification and decision problem. The novelty is the integration of multi-signal verification with a security-regression-aware acceptance policy and evaluation of the resulting end-to-end safety trade-off. The core outcome is therefore not merely a patch, but an evidence-backed decision about whether a patch should be trusted.

3.7 Objectives

1. Quantify the frequency and types of security regressions introduced by AI-generated fixes.
2. Build a multi-stage verification pipeline combining static, dynamic, semantic, and ML/LLM-based evidence.
3. Develop a confidence-based patch-acceptance mechanism.
4. Compare direct LLM repair with verification-driven repair under identical vulnerabilities and models.
5. Release reproducible benchmarks, evaluation scripts, and representative failure cases.

3.8 Research Timeline

Period	Planned Work
Month 1	Literature review; select vulnerability datasets and repositories; establish CodeQL/Bandit and direct-LLM repair baselines.
Month 2	Characterize baseline repair failures, functional breakage, and security regressions.
Months 3–4	Design and implement candidate generation, multi-signal verification, and confidence-based patch acceptance.
Month 5	Conduct cross-model and cross-language experiments, ablations, and real-repository validation.
Month 6	Analyze results, prepare reproducibility package and benchmark, and finalize manuscript for submission.

Proposal 4 Cost-Aware Autonomous Data Acquisition for Efficient ML

4.1 Proposed Title

Cost-Aware Autonomous Data Acquisition for Efficient and Adaptive Machine Learning

4.2 One-Line Summary

An intelligent agent that decides not only which data should be acquired, but also whether to label, retrieve, generate, or otherwise acquire it, maximizing model improvement per unit of acquisition cost.

4.3 Keywords

Active learning; data acquisition; autonomous agents; information value; cost-sensitive learning; large language models; human-in-the-loop ML; synthetic data; adaptive machine learning.

4.4 Abstract

The performance of machine-learning systems depends strongly on the quantity, quality, and representativeness of training data. In practical environments, however, collecting additional data can be expensive because it may require human annotation, expert intervention, database retrieval, experimentation, or computational generation. Conventional active-learning methods primarily focus on selecting the most informative samples for labeling, while recent LLM-based active-learning approaches have expanded this setting to selection, annotation, and generation. This research proposes a cost-aware autonomous data-acquisition agent that treats data acquisition as a sequential decision-making problem and chooses among multiple acquisition actions according to their expected information value and resource cost.

At each iteration, the agent will analyze model uncertainty, data coverage, class imbalance, feature-space gaps, historical acquisition outcomes, and estimated action costs. Candidate actions may include requesting a human label, retrieving an additional real sample, generating a

synthetic sample, or targeting a previously underrepresented region of the feature space. The agent will estimate the expected improvement associated with each action and optimize learning progress per unit cost rather than accuracy alone.

Evaluation will use classification and regression datasets with controlled budgets and realistic constraints including limited annotation resources, class imbalance, and distribution shift. Baselines will include random sampling, uncertainty sampling, diversity-based active learning, and representative LLM-assisted active-learning strategies. Performance will be evaluated using predictive quality, cumulative acquisition cost, convergence rate, robustness, and performance per unit resource expenditure. The expected contribution is a general framework for autonomous information acquisition that enables ML systems to determine not only what additional information they need, but also the most economically efficient way of obtaining it.

4.5 Prior Work

Active learning has traditionally focused on selecting informative examples for labeling. A comprehensive ACL 2025 survey showed that LLM-based active learning has already broadened the paradigm to include selection, generation, and cost-effective annotation [11]. More broadly, the emerging concept of "Data Savvy Agents" argues for agents that can proactively acquire, process, and adapt data rather than passively consume fixed datasets [12]. These works establish the importance of LLMs and agents in the active-learning loop, but they leave room for a unified decision framework in which multiple acquisition mechanisms compete under explicit resource costs.

4.6 Research Gap and Proposed Novelty

The proposed research moves from sample selection to *acquisition-action selection*. The agent does not assume that every useful observation must be human-labeled. It can choose between human labeling, real-data retrieval, synthetic generation, or targeted acquisition and compare these alternatives through their expected value and cost. The novelty therefore lies in modeling heterogeneous acquisition actions within a common sequential decision process and evaluating the system using cost-normalized learning efficiency.

4.7 Objectives

1. Formalize data acquisition as a cost-sensitive sequential decision problem.
2. Develop an agent that evaluates heterogeneous acquisition actions.
3. Estimate expected information value from uncertainty, coverage, and model behavior.
4. Compare autonomous acquisition against standard active-learning baselines under fixed resource budgets.
5. Study robustness when data distributions change during acquisition.

4.8 Research Timeline

Period	Planned Work
Month 1	Literature review; formalize acquisition-cost model; select benchmark tasks and implement standard active-learning baselines.
Month 2	Implement uncertainty, diversity, coverage, and information-value estimators; design the acquisition simulator.
Months 3–4	Develop the autonomous acquisition policy across labeling, retrieval, generation, and targeted sampling actions.
Month 5	Evaluate across datasets and budgets; conduct ablations and distribution-shift experiments.
Month 6	Analyze cost–performance trade-offs, prepare benchmark/code release, and finalize manuscript for submission.

Proposal 5 Task-Adaptive Error-Correction Allocation for Quantum Machine Learning

5.1 Proposed Title

Task-Adaptive Error-Correction Allocation for Variational Quantum Machine Learning under NISQ Noise

5.2 One-Line Summary

A learning-aware framework that allocates limited error-correction and error-mitigation resources to the most noise-sensitive parts of a QML circuit, seeking the best trade-off between model accuracy, trainability, and quantum-resource overhead.

5.3 Keywords

Quantum machine learning; quantum error correction; quantum error detection; quantum error mitigation; variational quantum circuits; NISQ; noise-aware training; resource optimization; quantum hardware.

5.4 Abstract

Quantum machine learning is highly sensitive to noise in near-term quantum hardware. Quantum error correction can improve reliability, but full fault-tolerant implementations introduce substantial qubit, circuit-depth, measurement, and latency overheads. Recent work has demonstrated both error-detecting QML implementations and partial error-correction strategies, but the amount and placement of protection required by a QML circuit may differ across gates, layers, parameters, datasets, and noise conditions. This research proposes a task-adaptive error-correction allocation framework in which protection resources are assigned according to the measured sensitivity of the QML computation rather than applied uniformly across the circuit.

The framework will profile a variational quantum classifier or related parameterized quantum circuit and estimate how local gate noise, accumulated circuit noise, and parameter sensitivity affect the final learning objective and gradient signal. Based on this profile, the system will

selectively allocate error-detection, partial error correction, error mitigation, or additional verification to the most influential regions of the circuit. The study will compare uniform protection, no protection, fixed partial protection, and the proposed task-adaptive strategy under depolarizing, dephasing, thermal, and hardware-derived noise models.

The key evaluation will measure not only final classification accuracy but also trainability, gradient behavior, logical/physical error rates, circuit overhead, number of physical qubits, syndrome or measurement overhead, and robustness across changing noise levels. The expected contribution is a principled way to spend limited quantum-error resources where they matter most for a machine-learning objective. The work is positioned as a bridge between QML training dynamics and resource-aware quantum error correction, with a near-term emphasis on simulatable experiments and optional validation on available cloud quantum hardware.

5.5 Prior Work

Adermann, Suzuki, and Usman demonstrated a Variational Quantum Classifier using the [[4,2,2]] stabiliser code for quantum error detection and showed that ancilla noise can constrain the achievable QML accuracy [13]. Kang *et al.* proposed partial QEC for QML and identified a resource-efficient operating regime that avoids some of the overhead associated with full fault tolerance [14]. Meyer *et al.* introduced variational quantum error correction in which error-correction encodings are learned for specific noise structures [15]. Separately, an RL-based co-design proposal has argued for jointly adapting QML circuits and QEC parameters [16], while other work has studied noise-aware QNN training and hardware-adaptive robustness [17]. These results show that the intersection of QML and error correction is now established; therefore, a generic "QEC for QML" proposal would not be sufficiently novel.

5.6 Research Gap and Proposed Novelty

The proposed novelty is narrower and more experimentally testable: instead of designing a new universal QEC code or simply applying an existing code to a QML circuit, the project will study whether *task sensitivity* can guide where and how much protection is deployed. Protection becomes an allocation problem. The research will seek a Pareto frontier between QML performance and protection overhead and determine whether selective protection can match or exceed uniform protection at substantially lower resource cost. The hypothesis is that not every part of a variational circuit contributes equally to downstream learning quality, so uniform error protection may be inefficient.

5.7 Objectives

1. Quantify the relationship between circuit-local noise sensitivity and QML training performance.
2. Develop a sensitivity-driven mechanism for allocating error-detection/correction resources.
3. Compare uniform, fixed, and task-adaptive protection strategies under multiple noise models.
4. Measure the accuracy–trainability–resource trade-off of each strategy.
5. Validate the framework in simulation and, where hardware access permits, on real quantum processors.

5.8 Research Timeline

Period	Planned Work
Month 1	Literature review; establish QML and quantum-error fundamentals; implement baseline VQC and reproducible noise models.
Month 2	Reproduce selected error-detection/partial-QEC results; establish accuracy, gradient, and resource-overhead baselines.
Months 3–4	Develop sensitivity profiling and the task-adaptive error-protection allocation mechanism; run initial simulations.
Month 5	Conduct cross-noise, cross-circuit, and cross-dataset experiments with ablation and resource-scaling studies.
Month 6	Consolidate the accuracy–trainability–resource Pareto analysis; optional hardware validation; prepare reproducibility package and manuscript.

References

[1] R. Ranjan, V. Hudovernik, M. Znidar, C. Kanatsoulis, R. Upendra, M. Mohammadi, J. Meyer, T. Palczewski, C. Guestrin, and J. Leskovec, "Relational Transformer: Toward Zero-Shot Foundation Models for Relational Data," in *International Conference on Learning Representations (ICLR)*, 2026. Available: https://proceedings.iclr.cc/paper_files/paper/2026/hash/7107d4d2e837bde2171c6b71b5bde954-Abstract-Conference.html

[2] Y. Wang, J. You, C. Shi, and M. Zhang, "Relational In-Context Learning via Synthetic Pre-training with Structural Prior," in *Proceedings of the 43rd International Conference on Machine Learning*, PMLR 306, 2026, pp. 131921–131962. Available: https://proceedings.mlr.press/v306/wang26jz.html

[3] D. O. Correa and F. G. Azevedo, "Context Window Failures in Relational Foundation Models," arXiv:2609.00460, 2026. Available: https://arxiv.org/abs/2609.00460

[4] R. Shi, Y. Wang, M. Du, X. Shen, and X. Wang, "A Comprehensive Survey of Synthetic Tabular Data Generation," arXiv:2504.16506, 2025. Available: https://arxiv.org/abs/2504.16506

[5] R. Challagundla, M. Dorodchi, P. Wang, and M. Lee, "Synthetic Tabular Data Generation: A Comparative Survey for Modern Techniques," arXiv:2507.11590, 2025. Available: https://arxiv.org/abs/2507.11590

[6] "Tabular data generation models: An in-depth survey and performance benchmarks with extensive tuning," *Neurocomputing*, vol. 658, art. 131655, 2025. DOI: https://doi.org/10.1016/j.neucom.2025.131655

[7] N. Nafis, I. Esnaola, A. Martinez-Perez, M.-C. Villa-Uriol, and V. Osmani, "Critical Challenges and Guidelines in Evaluating Synthetic Tabular Data: A Systematic Review," arXiv:2504.18544, 2025. Available: https://arxiv.org/abs/2504.18544

[8] M. Fu, C. K. Tantithamthavorn, V. Nguyen, and T. Le, "GPT for Vulnerability Detection, Classification, and Repair: How Far Are We?" in *Proceedings of the Asia-Pacific Software*

Engineering Conference (APSEC), 2023, pp. 632–636. DOI: https://doi.org/10.1109/APSEC60848.2023.00085

[9] V. Belozerov, P. J. Barclay, and A. Sami, "Secure coding with AI – from detection to repair," *Empirical Software Engineering*, vol. 31, art. 93, 2026. DOI: https://doi.org/10.1007/s10664-026-10812-8

[10] M. Surikov, "Securing AI-Generated Code: A Just-in-Time Vulnerability Detection and Remediation Pipeline," arXiv:2608.16187, 2026. Available: https://arxiv.org/abs/2608.16187

[11] Y. Xia et al., "From Selection to Generation: A Survey of LLM-based Active Learning," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025, pp. 14552–14569. DOI: https://doi.org/10.18653/v1/2025.acl-long.708

[12] N. Seedat, J. Liu, and M. van der Schaar, "What's the next frontier for Data-centric AI? Data Savvy Agents," arXiv:2511.01015, 2025. Available: https://arxiv.org/abs/2511.01015

[13] E. Adermann, H. Suzuki, and M. Usman, "Variational quantum machine learning with quantum error detection," *Quantum Machine Intelligence*, vol. 8, art. 1, 2026. DOI: https://doi.org/10.1007/s42484-026-00347-y

[14] H. Kang, Y. Kim, E. Adermann, M. Sevior, and M. Usman, "Almost fault-tolerant quantum machine learning with drastic overhead reduction," *Quantum Science and Technology*, vol. 11, no. 1, art. 015021, 2026. DOI: https://doi.org/10.1088/2058-9565/ae2157

[15] N. Meyer, C. Mutschler, A. Maier, and D. D. Scherer, "Learning Encodings by Maximizing State Distinguishability: Variational Quantum Error Correction," arXiv:2506.11552, 2025. Available: https://arxiv.org/abs/2506.11552

[16] S. H. Maes, "Adaptive Co-Design of Quantum Machine Learning Algorithms and Error Correction Protocols using Reinforcement Learning," 2025. Available: https://zenodo.org/records/15428358

[17] H. Wang, J. Gu, Y. Ding, Z. Li, F. T. Chong, D. Z. Pan, and S. Han, "QuantumNAT: Quantum Noise-Aware Training with Noise Injection, Quantization and Normalization," in *Proceedings of the International Conference on Computer-Aided Design (ICCAD)*, 2022/2023. Available: https://hanlab.mit.edu/projects/quantumnat